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STACKED SENSOR DEVICE

Abstract

A stacked sensor device includes a micro electromechanical system, MEMS, pressure sensor including a pressure sensor substrate having a recess formed therein, a flexible MEMS structure covering the recess, thereby forming a hermetic chamber within the pressure sensor substrate, and pressure sensing means for detecting a deflection of the flexible MEMS structure. The stacked sensor device further includes a gas sensor including gas sensing means for detecting a property of an ambient gas, the gas sensor being arranged on a standoff above the pressure sensor, such that a cavity is formed by the pressure sensor, the standoff and the gas sensor, and an opening that couples the cavity to the ambient gas.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Germany Patent Application No. 102024201710.8 filed on Feb. 23, 2024, the content of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to a stacked sensor device comprising a MEMS pressure sensor and a gas sensor as well as to a method of forming a stacked sensor device.

BACKGROUND

[0003] In recent years, advancements in sensor technology have paved the way for integrated devices that provide enhanced capabilities for monitoring environmental conditions. Until recently, integrated sensors have typically operated in isolation, thus limiting the ability to capture a holistic picture of environmental changes. By combining the strengths of a gas sensor, e.g., a humidity sensor, and a pressure sensor within a single, integrated framework, both an improved accuracy and an expansion of the range of potential applications of gas sensing is achieved. The range of applications include gas analysis, battery monitoring, process monitoring, human comfort measurement, or heating, ventilating, and air conditioning (HVAC) applications. Conventional integrated devices that include a humidity sensor and a pressure sensor typically feature a side-by-side arrangement of the sensing elements on the chip. However, such an arrangement easily clashes with the increasing demand of small-footprint system-in-package solutions. A further disadvantage is the cost-inefficient manner of such a side-by-side arrangement as chip surface is expensive.

SUMMARY

[0004] In some implementations, a stacked sensor device includes a micro electromechanical system, MEMS, pressure sensor that includes a pressure sensor substrate having a recess formed therein, a flexible MEMS structure covering the recess, thereby forming a hermetic chamber within the pressure sensor substrate, and pressure sensing means for detecting a deflection of the flexible MEMS structure. The stacked sensor device in these implementations further include a gas sensor that includes gas sensing means for detecting a property of an ambient gas around the stacked sensor device. The gas sensor is arranged on a standoff above the pressure sensor, such that a cavity is formed by the pressure sensor, the standoff, and the gas sensor. Furthermore, the stacked sensor device has an opening that couples the cavity to the ambient gas.

[0005] In some implementations, a method of forming a stacked sensor device includes providing a micro electromechanical system, MEMS, pressure sensor that includes a pressure sensor substrate having a recess formed therein, a flexible MEMS structure covering the recess, thereby forming a hermetic chamber within the pressure sensor substrate, and pressure sensing means for detecting a deflection of the flexible MEMS structure. The method further includes arranging a standoff on a top surface of the MEMS pressure sensor, arranging a gas sensor on a top surface of the standoff facing away from the MEMS pressure sensor, wherein the gas sensor includes gas sensing means for detecting a property of an ambient gas. The method further includes forming an opening, wherein the pressure sensor, the standoff and the gas sensor form a cavity, and the opening couples the cavity to the ambient gas.

[0006] Those skilled in the art will recognize additional features and advantages upon reading the following detailed description, and upon viewing the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The present disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like reference numerals refer to similar or identical elements. The elements of the drawings are not necessarily to scale relative to each other. The features of the various illustrated examples can be combined unless they exclude each other.

[0008] FIG. 1 illustrates a cross-sectional view of a first example implementation of a stacked sensor device.

[0009] FIG. 2 illustrates a cross-sectional view of a second example implementation of a stacked sensor device.

[0010] FIG. 3 illustrates a cross-sectional view of a third example implementation of a stacked sensor device.

[0011] FIG. 4 illustrates a cross-sectional view of a fourth example implementation of a stacked sensor device.

[0012] FIG. 5 illustrates a top view of an example implementation of a humidity sensor of a stacked sensor device.

DETAILED DESCRIPTION

[0013] Various examples will now be described more fully with reference to the accompanying drawings in which some examples are illustrated. In the figures, the thicknesses of lines, layers and/or regions may be exaggerated for clarity.

[0014] Accordingly, while further examples are capable of various modifications and alternative forms, some particular examples thereof are shown in the figures and will subsequently be described in detail. However, this detailed description does not limit further examples to the particular forms described. Further examples may cover all modifications, equivalents, and alternatives falling within the scope of the disclosure. Like numbers refer to like or similar elements throughout the description of the figures, which may be implemented identically or in modified form when compared to one another while providing for the same or a similar functionality.

[0015] It will be understood that when an element is referred to as being “connected” or “coupled” to another element, the elements may be directly connected or coupled or via one or more intervening elements. If two elements A and B are combined using an “or”, this is to be understood to disclose all possible combinations, e.g., only A, only B as well as A and B. An alternative wording for the same combinations is “at least one of A and B”. The same applies for combinations of more than 2 elements.

[0016] The terminology used herein for the purpose of describing particular examples is not intended to be limiting for further examples. Whenever a singular form such as “a,” “an” and “the” is used and using only a single element is neither explicitly or implicitly defined as being mandatory, further examples may also use plural elements to implement the same functionality. Likewise, when a functionality is subsequently described as being implemented using multiple elements, further examples may implement the same functionality using a single element or processing entity. It will be further understood that the terms “comprises,” “comprising,” “includes” and/or “including,” when used, specify the presence of the stated features, integers, steps, operations, processes, acts, elements and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, processes, acts, elements, components and/or any group thereof.

[0017] Unless otherwise defined, all terms (including technical and scientific terms) are used herein in their ordinary meaning of the art to which the examples belong.

[0018] FIG. 1 illustrates a cross-sectional view of a first example implementation of a stacked sensor device **10**. The stacked sensor device **10** includes a micro electromechanical system, MEMS, pressure sensor **20** comprising a pressure sensor substrate **21** formed from a semiconductor

material, e.g., silicon. A recess **22** is formed within the pressure sensor substrate **21**. The pressure sensor **20** further comprises a flexible MEMS structure **23** arranged on a surface of the pressure sensor substrate **21** such that the recess **22** is covered, in particular the recess **22** is completely covered, by the flexible MEMS structure **23**. The flexible MEMS structure for example is a diaphragm or a membrane formed from a semiconductor material, e.g., silicon or silicon nitride. Thus, the covered recess **22** forms a hermetic chamber **24** that is sealed from an environment of the pressure sensor **20**, in particular from the cavity **50**. The pressure sensor **20** further comprises pressure sensing means **25** for detecting a deflection of the flexible MEMS structure **23**. For example, the pressure sensing means **25** comprise electrodes arranged on opposite sides of the flexible MEMS structure **23**. For example, a capacitance between the electrodes depends on a deflection of the flexible MEMS structure **23**. The deflection of the flexible MEMS structure **23** in turn can depend on a pressure difference between a pressure of a vacuum or of a gas within the hermetic chamber **24** and a gas pressure outside the hermetic chamber **24**, e.g., a gas pressure of a gas within the cavity **50**. The working principle of MEMS pressure sensors is a well-established concept and is not further detailed throughout this disclosure.

[0019] The stacked sensor device **10** further comprises one or multiple standoffs **40** that are arranged on a top surface of the pressure sensor **20**. For example, the standoffs are arranged on a top layer of the pressure sensor, which can be the flexible MEMS structure **23**, for instance, as illustrated. Alternatively, the standoffs **40** can be arranged on surfaces of the pressure sensor substrate **21** that are not covered by the layer comprising the flexible MEMS structure **23**. For example, the standoffs **40** are formed from a semiconductor material that can be the same material as a material of the pressure sensor substrate **21**, e.g., silicon.

[0020] The stacked sensor device **10** further comprise a gas sensor **30** that is arranged on a surface of the standoffs **40** facing away from the pressure sensor **20**. In other words, the pressure sensor **20**, the standoffs **40** and the humidity sensor **30** form a stacked structure, e.g., the stacked sensor device **10**. The humidity sensor comprises a gas sensor substrate **31**, which is formed from a semiconductor material that can be the same material as a material of the pressure sensor substrate **21** and/or of the standoffs **40**, e.g., silicon. The gas sensor **30** further includes in this example implementation a sensing layer **39** arranged on a top surface of the gas sensor substrate **31**, wherein the sensing layer **39** comprises gas sensing means **35** (e.g., a gas sensing circuit), e.g., an electrode structure comprising two electrodes for measuring a capacitance between the electrodes, for instance. For example, the gas sensor **30** is a humidity sensor **30**, the gas sensor substrate **31** is a humidity sensor substrate **31**, and the gas sensing means **35** are humidity sensing means **35** (e.g., a humidity sensing circuit, or a gas sensing circuit with a humidity sensing circuit). Alternatively, the gas sensor can be a different type such as a temperature sensor or a chemicapacitive sensor for volatile organic compounds, for instance.

[0021] Moreover, in this example implementation, a gas sensitive element **36** is arranged within the sensing layer **39** in voids between the humidity sensing means **35**. The gas sensitive material **36** can be configured to adsorb or absorb water from the ambient gas and can have a dielectric property that depends on an amount of water adsorbed and/or absorbed by the gas sensitive element **36**. For example, the gas sensitive element **36** can be formed from a hygroscopic dielectric material, e.g., a plastic, a polymer such as a polyimide, or an oxide such as silicon oxide, with a dielectric constant that is proportional to an amount of water absorbed by the gas sensitive element **36**. Typically, at equilibrium conditions, the amount of moisture present in a hygroscopic material depends on both ambient temperature and ambient water vapor pressure. Therefore, absorption of moisture using the gas sensitive element **36** of the humidity sensor **30** results in an increase in a capacitance measured across a pair of electrodes forming the humidity sensing means **35** placed around the gas sensitive element **36** as described and illustrated. The working principle of capacitive humidity sensors is a well-established concept and is not further detailed throughout this disclosure. Alternatively to a capacitive humidity sensor, the humidity sensor **30** can be configured

as a resistive, e.g., piezoresistive, humidity sensor.

[0022] The pressure sensor **20**, the standoffs **40** and the humidity sensor form a cavity **50**. In other words, stacked sensor device **10** comprises a cavity **50** that is delimited by the pressure sensor **20**, e.g., by the flexible MEMS structure **23**, the standoff **40**, and the humidity sensor **30**, e.g., by the humidity sensor substrate **31**. The stacked sensor device **10** further comprises one or multiple openings **32** that couple the cavity **50** to an ambient gas surrounding the stacked sensor device **10**, for instance. The one or multiple openings **32** can be the only coupling between the cavity **50** and the ambient gas. For example, the stacked sensor device **32** comprises one or multiple openings **32** that extend from a top surface of the humidity sensor **30** to the cavity **50** as illustrated.

Alternatively or in addition, an opening **32** can extend through the standoff **40**. The cavity **50** can be configured to thermally decouple the pressure sensor **20** from the humidity sensor **30**.

Furthermore, the one or multiple openings **32** each can be dimensioned that gas molecules of an analyte gas, e.g., hydrogen or air, can enter the cavity **50**, while larger particles, e.g., contaminants, are prevented from entering the cavity **50**. Thus, the forming of a cavity **50** can act as a particle protection for the pressure sensor **20**. Moreover, the humidity sensor **30** can act as integrated electromagnetic compatibility (EMC) shielding for the pressure sensor **20**.

[0023] The stacked sensor device **10** may further comprise active semiconductor components that may be part of an ASIC. The ASIC may be used for acquiring and processing electronic measurement signals from the pressure sensor **20** and the humidity sensor **30** and for generating pressure and humidity signals from the acquired measurement signals.

[0024] FIG. 2 illustrates a cross-sectional view of a second example implementation of a stacked sensor device **10**. In this implementation, the pressure sensing means **25** (e.g., a pressure sensing circuit) of the pressure sensor include a first electrode **26** arranged at a bottom side of the recess **22** opposite the flexible MEMS structure **23**, and a second electrode **27** arranged on a side of the flexible MEMS structure **23** facing the recess **22**. In other words, the first and second electrodes are arranged inside the hermetic chamber **24** and opposite each other in a vertical direction with respect to a main plane of extension of the pressure sensor **20**. Thus, a capacitance measured across the first and second electrodes **26**, **27** can depend on a deflection of the flexible MEMS structure **23** and hence on a gas pressure of gas inside the cavity **50**.

[0025] Moreover, in this second example implementation, the humidity sensor **30** comprises a capping layer **39a** that acts as the gas sensitive element **36**. Thus, the capping layer **39a** can be at least in portions formed from a hygroscopic dielectric material, e.g., a plastic, a polymer such as a polyimide, or an oxide such as silicon oxide, with a dielectric constant that is proportional to an amount of water absorbed by the capping layer **39a**. Underneath the capping layer **39a**, e.g., in between the humidity sensor substrate **31** and the capping layer **39a**, an electrode layer **39b** is arranged comprising a first electrode structure **33** and a second electrode structure **34** as the humidity sensing means **35** (e.g., a humidity sensing circuit, or a gas sensing circuit with a humidity sensing circuit). For example, the first and second electrode structures **33**, **34** form an interdigitated electrode structure, across which a capacitance can be measured that depends on an amount of water adsorbed and/or absorbed by the gas sensitive element **36** of the capping layer **39a**. The first and second electrode structures **33**, **34** can be separated by voids or an insulating material, e.g., silicon or silicon oxide. Furthermore, in between the electrode layer **39b** and the capping layer **39a** and/or in between the electrode layer and the humidity sensor substrate **31** an additional passivation layer can be arranged, e.g., a thin layer formed from silicon oxide or other suitable materials.

[0026] The opening **32** in this example implementation extends from the cavity **50** in a vertical direction through the humidity sensor substrate **31**, the electrode layer **39b**, and the capping layer **39a** as well as through the optional additional passivation layers.

[0027] FIG. 3 illustrates a cross-sectional view of a third example implementation of a stacked sensor device **10**. In contrast to the second implementation, in this third implementation the stacked

sensor device **10** further comprises a heater structure **37** arranged in proximity to the humidity sensing means **35** and to the gas sensitive element **36**. For example, the heater structure **37** is arranged within the humidity sensor substrate **31** as illustrated. Alternatively, the heater structure **37** can be arranged in an intermediate layer arranged in between the humidity sensor substrate **31** and a layer comprising the humidity sensing means **35** and/or the gas sensitive element **36**. The heater structure **37** can be formed from resistive heating elements, for example. The heater structure **37** is operable to heat the gas sensitive element **36**, e.g., for reconditioning purposes. In other words, the heater structure **37** is operable to heat the gas sensitive element **36** to evaporate adsorbed or absorbed water from the gas sensitive element **36**. This heating could be performed in between humidity measurements. To this end, the stacked sensor device **10** can further comprise active semiconductor components that may be part of an ASIC. The ASIC may be used for activating, deactivating and controlling the heater structure **37** and for thermalizing the gas sensitive element **36** to a target temperature. The humidity sensor **30** for temperature stabilization can further comprise a thermometer, e.g., a thermistor, arranged in proximity to the gas sensitive element **36**. [0028] In implementations, in which the humidity sensor **30** comprises the heater structure **37**, the cavity **50** in addition to EMC shielding and particle protection also thermally decouples the humidity sensor **30** from the pressure sensor **20**. This enables power-efficient usage of the sensor also under harsh and high-humidity conditions.

[0029] FIG. **4** illustrates a cross-sectional view of a fourth example implementation of a stacked sensor device **10**. In contrast to the third implementation, the humidity sensor **30** comprises a stacked structure **38** formed by a first layer comprising the first electrode structure **33**, a second layer comprising the gas sensitive element **36**, and a third layer comprising the second electrode structure **34**, the stacked structure **38** realizing a sandwich structure extending in a perpendicular direction with respect to a main plane of extension of the humidity sensor **30**. Optional passivation layers, for instance, can be arranged in between the first, second and third layers of the stacked structure **38** and/or in between the humidity sensor substrate **31** and the stacked structure **38**. Furthermore, a capping layer **39a** can be arranged on a top surface of the stacked structure **38** facing away from the humidity sensor substrate **31**.

[0030] FIG. **5** illustrates a top view of an example implementation of a humidity sensor **30** of a stacked sensor device **10**. The implementation shows the humidity sensor substrate **31** comprising multiple openings **32** towards the cavity **50**, the heater structure **37** arranged within or on a top surface of the humidity sensor substrate **31**, and the humidity sensing means **35** formed by a plurality of parallel pairs of first electrode structures **33** and second electrode structures **34**. For illustration purposes, the gas-sensitive element **36**, for example arranged within a capping layer **39a** on a top surface of the electrode layer **38b** comprising the first and second electrode structures **33**, **34**, is merely indicated in the Figure.

[0031] FIGS. **1** to **5** show only stacked sensor devices **10** having single sets of a pressure sensor **20** and a humidity sensor **30**. However, several sets of stacked sensor devices **10** may be combined to form an array of sets of stacked sensor devices **10**. In addition, each stacked sensor devices **10** may comprise more than one pressure sensor **20** and/or more than one humidity sensor **30**.

[0032] The combination of a particle and EMC protected pressure sensor **20** and humidity sensor **30**, the described stacked sensor device **10** brings the advantage of cost-efficient, 3D integration for a system-in-package. With the cavity **50** being arranged in between the optional heater structure **31** and the pressure sensor **20**, a power-efficient usage of the sensor even under harsh and high humidity conditions is enabled. The geometry and arrangement of the example pressure and humidity sensing means **25**, **35**, the heater structure **37** can be adapted to optimize for example sensitivity and/or robustness.

[0033] Although specific examples have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific examples shown and described without

departing from the scope of the present implementation. This application is intended to cover any adaptations or variations of the specific examples discussed herein. Therefore, it is intended that this implementation be limited only by the claims and the equivalents thereof.

[0034] It should be noted that the methods and devices including its preferred implementations as outlined in the present document may be used stand-alone or in combination with the other methods and devices disclosed in this document. In addition, the features outlined in the context of a device are also applicable to a corresponding method, and vice versa. Furthermore, all aspects of the methods and devices outlined in the present document may be arbitrarily combined. In particular, the features of the claims may be combined with one another in an arbitrary manner.

[0035] It should be noted that the description and drawings merely illustrate the principles of the proposed methods and systems. Those skilled in the art will be able to implement various arrangements that, although not explicitly described or shown herein, embody the principles of the implementation and are included within its spirit and scope. Furthermore, all examples and implementations outlined in the present document are principally intended expressly to be only for explanatory purposes to help the reader in understanding the principles of the proposed methods and systems. Furthermore, all statements herein providing principles, aspects, and implementations of the implementation, as well as specific examples thereof, are intended to encompass equivalents thereof.

ASPECTS

[0036] In particular, the following aspects are disclosed:

[0037] Aspect 1: A stacked sensor device (10), comprising a micro electromechanical system, MEMS, pressure sensor (20) including a pressure sensor substrate (21) having a recess (22) formed therein, a flexible MEMS structure (23) covering the recess (22), thereby forming a hermetic chamber (24) within the pressure sensor substrate (21), and pressure sensing means (25) for detecting a deflection of the flexible MEMS structure (23). The stacked sensor device (10) further comprises a gas sensor (30) including gas sensing means (35) for detecting a property of an ambient gas, the gas sensor (30) being arranged on a standoff (40) above the pressure sensor (20), such that a cavity (50) is formed by the pressure sensor (20), the standoff (40) and the gas sensor (30), and an opening (32) that couples the cavity (50) to the ambient gas.

[0038] Aspect 2: The stacked sensor device (10) according to aspect 1, wherein the opening (32) extends from a top surface of the gas sensor (30) to the cavity (50).

[0039] Aspect 3: The stacked sensor device (10) according to aspect 1 or 2, wherein the cavity (50) thermally decouples the pressure sensor (20) from the gas sensor (30).

[0040] Aspect 4: The stacked sensor device (10) according to one of aspects 1 to 3, wherein the opening (32) is dimensioned to allow gas molecules to enter the cavity (50), and to prevent contaminating particles from entering the cavity (50).

[0041] Aspect 5: The stacked sensor device (10) according to one of aspects 1 to 4, wherein the gas sensor (30) comprises a gas sensor substrate (31), and wherein the standoff (40), the pressure sensor substrate (21) and the gas sensor substrate (31) are formed from the same material.

[0042] Aspect 6: The stacked sensor device (10) according to one of aspects 1 to 5, wherein the gas sensor (30) is a capacitive or resistive gas sensor.

[0043] Aspect 7: The stacked sensor device (10) according to one of aspects 1 to 6, wherein the gas sensing means (35) comprise a first electrode structure (33), a second electrode structure (34) and a gas sensitive element (36), the gas sensitive element (36) being configured to adsorb water from the ambient gas and having a dielectric property that depends on an amount of water adsorbed by the gas sensitive element (36).

[0044] Aspect 8: The stacked sensor device (10) according to aspect 7, wherein the gas sensor (30) further comprises a heater structure (37).

[0045] Aspect 9: The stacked sensor device (10) according to aspect 8, wherein the heater structure (37) is operable to heat the gas sensitive element (36).

[0046] Aspect 10: The stacked sensor device (10) according to aspect 8 or 9, wherein the heater structure (37) is arranged between the cavity (50) and the gas sensitive element (36).

[0047] Aspect 11: The stacked sensor device (10) according to one of aspects 7 to 10, wherein a capacitance between the first electrode structure (33) and the second electrode structure (34) depends on the dielectric property of the gas sensitive element (36).

[0048] Aspect 12: The stacked sensor device (10) according to one of aspects 7 to 11, wherein the first electrode structure (33), the gas sensitive element (36) and the second electrode structure (34) form a stacked structure (38) extending in a perpendicular direction with respect to a main plane of extension of the gas sensor (30).

[0049] Aspect 13: The stacked sensor device (10) according to one of aspects 7 to 11, wherein the first electrode structure (33) and the second electrode structure (34) are arranged in a sensing layer (39) with the gas sensitive element (36) being arranged within the sensing layer (39) in between the first electrode structure (33) and the second electrode structure (34).

[0050] Aspect 14: The stacked sensor device (10) according to one of aspects 7 to 11, wherein the first electrode structure (33) and the second electrode structure (34) form interdigitated electrodes with the gas sensitive element (36) arranged in between and/or on the interdigitated electrodes.

[0051] Aspect 15: The stacked sensor device (10) according to one of aspects 7 to 14, wherein the gas sensitive element (36) is formed from an oxide, in particular from a silicon oxide, or from a polymer, in particular from a polyimide.

[0052] Aspect 16: The stacked sensor device (10) according to one of aspects 7 to 15, wherein the gas sensor (30) further comprises a capping layer (39a) arranged on the first and second electrode structures (33, 34).

[0053] Aspect 17: The stacked sensor device (10) according to aspect 16, wherein the capping layer (39a) is configured as the gas sensitive element (36).

[0054] Aspect 18: The stacked sensor device (10) according to one of aspects 1 to 17, wherein the flexible MEMS structure (23) is a membrane or a diaphragm.

[0055] Aspect 19: The stacked sensor device (10) according to one of aspects 1 to 18, wherein the pressure sensing means (25) of the pressure sensor (20) comprise a first electrode (26) arranged at a bottom side of the recess (22) opposite the flexible MEMS structure (23), and a second electrode (27) arranged on a side of the flexible MEMS structure (23) facing the recess (22).

[0056] Aspect 20: The stacked sensor device (10) according to one of aspects 1 to 19, wherein the gas sensor (30) is a humidity sensor having humidity sensing means as the gas sensing means (35), the humidity sensor being configured to detect a humidity level in the ambient gas as the property of the ambient gas.

[0057] Aspect 21: A method of forming a stacked sensor device (10), the method comprising: providing a micro electromechanical system, MEMS, pressure sensor (20), the MEMS pressure sensor (20) comprising a pressure sensor substrate (21) having a recess (22) formed therein, a flexible MEMS structure (23) covering the recess (22), thereby forming a hermetic chamber (24) within the pressure sensor substrate (21), and pressure sensing means (25) for detecting a deflection of the flexible MEMS structure (23). The method further comprises arranging a standoff (40) on a top surface of the MEMS pressure sensor (20), arranging a gas sensor (30) on a top surface of the standoff (40) facing away from the MEMS pressure sensor (20), the gas sensor (30) comprising gas sensing means (35) for detecting a property of an ambient gas, and forming an opening (32), wherein the pressure sensor (22), the standoff (40) and the humidity sensor (30) form a cavity (50), and the opening (32) couples the cavity (50) to the ambient gas.

[0058] Aspect 22: The method according to claim 21, wherein arranging the gas sensor (30) comprises arranging a humidity sensor on a top surface of the standoff (40).

Claims

- 1.** A stacked sensor device, comprising: a micro-electromechanical system (MEMS) pressure sensor comprising: a pressure sensor substrate having a recess formed in the pressure sensor substrate; a flexible MEMS structure covering the recess, thereby forming a hermetic chamber within the pressure sensor substrate; and a pressure sensing circuit for detecting a deflection of the flexible MEMS structure; a gas sensor comprising a gas sensing circuit for detecting a property of an ambient gas, the gas sensor being arranged on a standoff above the MEMS pressure sensor, such that a cavity is formed by the MEMS pressure sensor, the standoff, and the gas sensor; and an opening that couples the cavity to the ambient gas.
- 2.** The stacked sensor device according to claim 1, wherein the gas sensor is a humidity sensor having a humidity sensing circuit as the gas sensing circuit, the humidity sensor being configured to detect a humidity level in the ambient gas as the property of the ambient gas.
- 3.** The stacked sensor device according to claim 1, wherein the opening extends from a top surface of the gas sensor to the cavity.
- 4.** The stacked sensor device according to claim 1, wherein the cavity thermally decouples the MEMS pressure sensor from the gas sensor.
- 5.** The stacked sensor device according to claim 1, wherein the opening is dimensioned to allow gas molecules to enter the cavity, and to prevent contaminating particles from entering the cavity.
- 6.** The stacked sensor device according to claim 1, wherein the gas sensor comprises a gas sensor substrate, and wherein the standoff, the pressure sensor substrate, and the gas sensor substrate are formed from the same material.
- 7.** The stacked sensor device according to claim 1, wherein the gas sensor is a capacitive gas sensor or a resistive gas sensor.
- 8.** The stacked sensor device according to claim 1, wherein the gas sensing circuit comprise a first electrode structure, a second electrode structure, and a gas sensitive element, the gas sensitive element being configured to adsorb water from the ambient gas and having a dielectric property that depends on an amount of water adsorbed by the gas sensitive element.
- 9.** The stacked sensor device according to claim 8, wherein the gas sensor further comprises a heater structure.
- 10.** The stacked sensor device according to claim 9, wherein the heater structure is operable to heat the gas sensitive element.
- 11.** The stacked sensor device according to claim 9, wherein the heater structure is arranged between the cavity and the gas sensitive element.
- 12.** The stacked sensor device according to claim 8, wherein a capacitance between the first electrode structure and the second electrode structure depends on the dielectric property of the gas sensitive element.
- 13.** The stacked sensor device according to claim 8, wherein the first electrode structure, the gas sensitive element, and the second electrode structure form a stacked structure extending in a perpendicular direction with respect to a main plane of extension of the gas sensor.
- 14.** The stacked sensor device according to claim 8, wherein the first electrode structure and the second electrode structure are arranged in a sensing layer, with the gas sensitive element being arranged within the sensing layer in between the first electrode structure and the second electrode structure.
- 15.** The stacked sensor device according to claim 8, wherein the first electrode structure and the second electrode structure form interdigitated electrodes, with the gas sensitive element being arranged in between or on the interdigitated electrodes.
- 16.** The stacked sensor device according to claim 8, wherein the gas sensitive element is formed from an oxide or from a polymer.
- 17.** The stacked sensor device according to claim 8, wherein the gas sensor further comprises a capping layer arranged on the first electrode structure and second electrode structure.

- 18.** The stacked sensor device according to claim 17, wherein the capping layer is configured as the gas sensitive element.
- 19.** The stacked sensor device according to claim 1, wherein the flexible MEMS structure is a membrane or a diaphragm.
- 20.** The stacked sensor device according to claim 1, wherein the pressure sensing circuit of the MEMS pressure sensor comprises: a first electrode arranged at a bottom side of the recess opposite the flexible MEMS structure; and a second electrode arranged on a side of the flexible MEMS structure facing the recess.
- 21.** A method of forming a stacked sensor device, the method comprising: providing a micro-electromechanical system (MEMS) pressure sensor, the MEMS pressure sensor comprising: a pressure sensor substrate having a recess formed in the pressure sensor substrate; a flexible MEMS structure covering the recess, thereby forming a hermetic chamber within the pressure sensor substrate; and a pressure sensing circuit for detecting a deflection of the flexible MEMS structure; arranging a standoff on a top surface of the MEMS pressure sensor; arranging a gas sensor on a top surface of the standoff facing away from the MEMS pressure sensor, the gas sensor comprising gas sensing circuit for detecting a property of an ambient gas; and forming an opening, wherein the MEMS pressure sensor, the standoff, and the gas sensor form a cavity, and wherein the opening couples the cavity to the ambient gas.
- 22.** The method according to claim 21, wherein arranging the gas sensor comprises arranging a humidity sensor on the top surface of the standoff.
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